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Analysis and Detection of Speech under Emotional Stress

Ján Staš, Daniel Hládek, Zuzana Sokolová, Marek Čech, Katarína Škotková, and Patrik Poremba
Department of Electronics and Multimedia Communications, Faculty of Electrical Engineering and Informatics,
Technical University of Košice, Němcovej 32, 042 00 Košice, Slovakia
jan.stas@tuke.sk, daniel.hladek@tuke.sk, zuzana.sokolova@tuke.sk,
marek.cech@student.tuke.sk, katarina.skotkova@student.tuke.sk, patrik.poremba@student.tuke.sk

***Abstract*—Automatic detection of emotional stress in speech has recently drawn increasing attention, mainly in the fields of natural and spoken language processing. Stress can be automatically detected from speech-derived parameters, such as fundamental frequency F0, voice intensity, jitter, shimmer, harmonic-to-noise ratio, mel-frequency cepstral coefficients, etc. In this paper, we focus on the analysis of selected voice characteristics in available datasets of emotional speech and speech under stress in the English and Slovak languages. We also proposed a stress level detection approach, which uses the fusion of the Teager energy operator (TEO) with mel-frequency cepstral coefficients (MFCCs) and classification based on decision trees and ensemble learning method of bagging decision trees. Preliminary results of 64% classification accuracy are comparable to other studies in this field of research.**

I. Introduction

The analysis and detection of emotional stress in human speech is currently one of the hot research topics in the field of natural language processing (NLP) and automatic speech recognition (ASR). With the increasing occurrence of stress in modern society and its negative effects on the health and performance of people, this topic is becoming more and more important.

Acute stress usually means large demands on an individual's physical and psychological resources, which are characterized by the release of stress hormones and other physiological and psychological changes, including changes in speech. Stress research is mostly focused on psychological aspects induced by mathematical tasks, exams, or driving, and physiological aspects, such as physical activities.

When creating datasets of recordings of spontaneous speech under stress, it is necessary to realize that speakers cannot be deliberately exposed to high-stress or dangerous situations. On the contrary, the use of speech recordings from emergency lines, aircraft communication systems, or military and rescue services is prevented by security aspects and GDPR.

Since the effect of stress in speech is non-specific and depends on the immediate situation and has a different temporal development than physiological indicators of stress (changes in skin conductivity, heart rate variability, in the content of stress hormones in the blood, and so on), it is extremely difficult to find a correlation between them for annotation of the exact stress level. Causing extreme levels of acute stress could be not only unethical but also health-threatening, so acted (simulated) datasets are often used. The development of systems for analyzing the speech of a stressed speaker is often limited to the small amount of data available, and advanced deep learning techniques cannot therefore be effectively applied.

Datasets of emotional speech provide slightly more data for training such systems. Strong emotions can also be considered as a kind of deviation from a neutral mental state, and they also affect speech and other physiological characteristics. However, imperfect evaluation of the intensity of emotion in speech during manual annotation may cause further complications in the research and development of the systems for detecting the presence of stress in speech.

The rest of the paper is organized as follows. In Section II, we briefly describe current approaches for detecting emotions and stress in speech. Section III summarizes the available datasets of emotional speech and speech under stress used in the experimental part of this research. In Section IV, we focused on the analysis of selected voice characteristics in the used databases, which clearly define the stress levels. The proposed stress level detection approach, which uses the fusion of the Teager energy operator with mel-frequency cepstral coefficients and classification based on decision trees and bagged trees, is presented in Section V. The preliminary experimental results of the proposed approach are also presented here. Finally, Section VI summarizes the contribution of our work and concludes the paper with future directions.

II. State-of-the-Art

Automatic speech emotion recognition has been researched since the early 1990s [1]. At the end of the 1990s, the first works aimed at detecting the presence of stress in speech were also published in [2].

Since a full overview of the approaches goes far beyond the scope of this paper, we summarized here only the latest approaches based on voice analysis of fundamental frequency F0 and classification using machine learning.

A comprehensive survey on stress detection through speech analysis using machine learning techniques is summarized in the paper by Vamsinath et al. [3].

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Wöllmer et al. [4] proposed an approach for continuous emotion recognition based on long short-term memory (LSTM) recurrent neural networks (RNNs) which include modeling of long-range dependencies between the acoustic observations derived from prosodic, spectral, and voice quality features. Stuhlsatz et al. [5] introduced generalized discriminant analysis (GerDA) based on deep neural networks (DNNs) for the task of acoustic emotion recognition. Mao et al. [6] proposed to learn affect-salient features for speech emotion recognition using convolutional neural networks (CNNs). The first end-to-end system for speech emotion recognition uses a combination of CNN with LSTM networks, in order to automatically learn the best representation of the speech signal directly from the raw time representation [7]. A number of emotional speech datasets in several languages are currently available to train and test these approaches. A comprehensive review of them is summarized in [8]. These include speech databases containing both acted and real emotions. Progress in this field was also supported by the organization of competitions such as the Valence and Arousal 'in-the-wild' Challenge [9].

The results of the study by Demenko and Jastrzebska [10] confirm the significance of the F0 for investigating stress. The shift in the F0 register constitutes the primary indicator of stress, especially when caused by fear. Also, a systematic increase in the range of F0 variability for the stress related to anger or irritation was observed. Van Puyvelde et al. [11] outlined the different stages of voice output, such as breathing, phonation, and resonance in relation to a neurovisceral integrated perspective on stress and human performance. They concluded that increased F0 values correspond to situations when cognitive control is lost, such as in life-threatening emergency situations (e.g., flight crashes, or alcohol intoxication). Decreased F0 values are consequential of high cognitive load. The mean F0 values give insight into arousal activity and the effort a subject can generate. Jitter expresses arousal in an inverse manner. Cognitive and emotional load correspond with respective small and larger reduced jitter values.

Han et al. [12] proposed a deep learning-based psychological stress detection algorithm from speech signals. The algorithm first extracts mel-frequency cepstral coefficients and then predicts the status of stress output using a binary decision criterion (no-stress or stress speech) using LSTM and feed-forward neural networks. The authors achieved 66.4% accuracy in predicting stress speech. A similar approach based on the extraction of MFCCs and predicting stress output using CNNs with dense fully connected layer networks was presented by Vaikole et al. [13]. Here the overall accuracy increased up to 94.33%. Both studies were performed using the RAVDESS dataset [14]. The research by Hilmy et al. [15] emphasizes how stress detection can be done by using the analysis of speech signals. The MFCCs were extracted from the speech signals and the stress was detected with the CNNs. It has been concluded that psychological stress could be detected through speech signals by analyzing the count of pauses and maximum amplitude, while the no-stress speech has a low count of pauses and maximum amplitude, stress speech has a high count of pauses and maximum amplitude. The authors achieved 67% accuracy in stress classification in real conditions of usage. Moreover, stressful situations can also be simulated with a computer game [16].

TABLE I.
STATISTICS OF SELECTED DATASETS

Dataset name	Type of speech	Recording scenario	Language	Number of speakers
RAVDESS	emotional speech	simulated	North American English	24 (12F, 12M)
SUS	emotional speech	simulated	Slovak	7 (4F, 3M)
CRISIS	expressive speech	simulated	Slovak	15
SUSAS	speech under stress	actual and simulated	English	32 (13F, 19M)
StressDat	speech under stress	simulated	Slovak	30 (16F, 14M)

III. AVAILABLE DATASETS OF EMOTIONAL SPEECH AND SPEECH UNDER STRESS

From several publicly available datasets of simulated (acted) and real (actual) stress and emotions, we have created one large emotional-stress speech dataset, with the help of which it will be possible to train automatic systems for detecting levels of emotional stress in human speech, to investigate the relationship between emotions and stress, but also to assess the possibility of measuring of real emotions and stress by using models trained on these datasets. In the following subsections, selected databases of emotional speech and speech under stress will be briefly described. Basic information about them is summarized in Table I.

A. RAVDESS

The RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song) is a multimodal dataset of emotional speech. The dataset consists of gender-balanced speech recordings obtained from 24 professional actors (12 females and 12 males), vocalizing lexically matched statements in a neutral North American English. Speech includes calm, happy, surprised, disgusted, sad, angry, and fearful expressions. Each expression is produced at a neutral level and two levels of emotional intensity. The set of 4,320 speech recordings was rated 10 times on emotional validity, intensity, and genuineness [14].

B. SUS

The SUS dataset (Courtroom sessions, "Súdna sieň" in Slovak) consists of recordings of emotional speech in the Slovak language of fabricated legal proceedings in which non-professional actors take part. Each session is oriented to a juridical case solution therefore emotional range of recorded utterances covers mostly emotions of neutral state and curiosity, and negative emotions, such as anger, aggressiveness, sadness, and disgust. Positive emotions are rare to find. All utterances were manually labeled. Emotional evaluation was provided on the whole sentences. In the case where more than one emotion occurred in an utterance, the division of such an utterance into shorter segments following the rule of information relevance was carried out. The whole sentences or segments were then evaluated from an emotional point of view (e.g. CU for curiosity, N for neutral, etc.). The SUS consists of approximately 2,000 utterances spanning from 5 to 6 seconds obtained from 7 speakers (4 females and 3 males) in different emotional states: neutral, curiosity, anger, sadness, and so on. All sessions were recorded with a 48 kHz sampling rate and 16-bit resolution [17].

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C. CRISIS

The CRISIS dataset contains high-quality recordings of expressive speech (under stress) realized in an acoustically adjusted recording studio, recorded with a 48 kHz sampling rate and 16-bit resolution. The authors designed a three-step method when creating the dataset. The speakers were instructed to pronounce the sentence once in a neutral manner (level 1 of tense arousal), then with higher imperativeness, as if it were a serious command or instruction (level 2), and finally like an extremely urgent command or statement made in a situation in which human lives are directly threatened (level 3). A similar scenario was also used when recording the lower tense arousal triplet of the database. After each session, the speakers relaxed for several seconds and then they started with a new sentence. The authors assume that this approach produces three emotionally consistent sets of utterances with three relatively distinguishable degrees of tense arousal. The CRISIS dataset contains 150 warning sentences for higher tense arousal (levels 1, 2, 3) and 150 sentences for lower tense arousal (levels -1, -2, -3) [18].

D. SUSAS

Generally, the best-known database of Speech Under Simulated and Actual Stress is SUSAS. The database consists of speech recordings with a broad range of emotions and stress levels. It contains over 16 thousand emotional speech utterances recorded under both acted and simulated stress collected from 32 English speakers (19 males and 13 females) in four different conditions: neutral stress, medium stress, high stress, and screaming. The time duration of audio samples spans from 1 to 2 seconds. Most of the utterances are one- or two-word commands, some contain background noise, and the recordings have a low sampling frequency (8 kHz) [2].

E. StressDat

The StressDat is a collection of acted speech recordings in the Slovak language, where actors uttered sentences in three different stress levels. The dataset is gender-balanced with 30 speakers (16 females and 14 males). Each participant was represented with various scenarios of potentially stressful situations and the context that manipulates the stress level from low over medium to high. Each scenario contained multiple related sentences and the subjects were asked to read all scenario sentences in the three stress levels. The dataset consists of a total of 12,813 utterances spanning from 6 to 31 seconds. The mean duration of utterances is 4.57 seconds. The dataset comprises a total of 79.30 hours [19].

IV. Analysis of Speech Characteristics

In order to be able to start training classifiers to detect stress levels in speech, it was first necessary to analyze individual databases from the point of view of basic speech parameters, such as fundamental frequency F0, jitter, shimmer, harmonic-to-noise ratio (HNR), and voice intensity. This analysis aimed to confirm findings from other studies on how individual parameters change with increasing or decreasing levels of stress.

Since we worked with emotional datasets RAVDESS and SUS, it was necessary to find a correlation between emotional states and stress levels and to unite them from the point of view of three basic stress levels (neutral, medium, and high stress) and only then could we start analyzing them. When assigning emotional states to individual stress levels, we used following characteristics, such as positive and negative stress, intensity, and timbre of the voice. The final mapping of emotional states to stress levels is shown in Table II. For each emotional state, the voice intensity parameter is also given (1 - low intensity, 2 - high intensity).

TABLE II.
Mapping emotion states to stress levels

Stress level	neutral	medium	high
RAVDESS	neutral (1,2) calm (1,2) happy (1) surprised (1) disgusted (1)	happy (2) surprised (2) disgusted (2) sad (1) fearful (1) angry (1)	sad (2) fearful (2) angry (2)
SUS	neutral (1,2) happy (1) surprised (1) disgusted (1)	happy (2) surprised (2) disgusted (2) sad (1)	sad (2) angry (1,2)

Table III. summarizes the average values of the parameters F0, jitter, shimmer, HNR, and voice intensity for all speakers in datasets at individual stress levels. In addition, we also created one large merged emotional-stress speech dataset (last column of Table III.) usable for training systems to detect the presence of stress in speech. However, it should be noted that we worked mainly with databases in which individual levels of emotion and stress in speech were simulated by actors (except for part of the SUSAS dataset). For this reason, the values of the parameters may not coincide with the values if they were speech recordings under actual stress.

Based on the values summarized in Table III, it can be concluded that as the fundamental frequency F0 and voice intensity increase, the level of stress increases, which agrees with the results of the research studies mentioned in Section II [10][11]. If we look at the values of the jitter, shimmer, and HNR parameters for individual databases, we cannot clearly demonstrate their influence on speech stress levels. However, if we look at the values in the last column of the Table III., for the merged dataset, the average values of the parameters jitter, shimmer, and HNR decrease with increasing stress levels. This confirms that the larger and more diverse the dataset of emotions and stress we have, the higher the accuracy of the resulting algorithm for detecting stress in speech can be. In addition, further increase in the size of the dataset could be achieved by data augmentation, for example by using generative adversarial networks (GAN).

V. Experiments

The experiments were focused on training and evaluating classifiers for the automatic prediction of stress levels on the data mentioned in Section III. Although the tool used offers several classification methods, such as Naïve Bayes (NB) classifier, logistic regression, support vector machines (SVM), k-nearest neighbor classifier (kNN), etc., in the experimental part we chose only two methods that produced the overall best results, namely standard decision trees and ensemble method of bagging decision trees.

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TABLE III.
EXTRACTED VOICE CHARACTERISTICS ACROSS SELECTED EMOTIONAL AND SPEECH DATASETS

Voice characteristic	Stress level	RAVDESS	SUS	CRISIS	SUSAS actual	SUSAS simulated	StressDat	Merged dataset
F0 [Hz]	neutral	180,8794	157,6620	173,8029	152,4761	118,6620	161,9615	161,7039
	medium	214,1468	165,0827	213,1216	159,0866	148,1999	189,0753	191,2940
	high	253,3393	178,4807	268,1447	198,0249	149,1937	228,4290	228,7341
jitter [%]	neutral	2,4106	2,3688	2,1980	1,8683	2,2472	2,4768	2,2623
	medium	2,4925	2,3423	1,8456	1,8419	2,3027	2,4250	2,1430
	high	2,4059	2,1981	1,6148	2,0303	2,4809	2,4472	2,1144
shimmer [dB]	neutral	0,1171	0,1021	0,0909	0,0691	0,0790	0,1066	0,0930
	medium	0,1195	0,0985	0,0881	0,0661	0,0749	0,1118	0,0926
	high	0,1215	0,0949	0,0835	0,0803	0,0749	0,1139	0,0925
HNR [dB]	neutral	9,9630	9,8181	12,6742	12,3903	11,1965	11,3231	11,6565
	medium	9,5225	10,1283	12,8073	11,8208	10,2817	10,7475	11,5824
	high	10,2873	10,5944	13,5091	10,2054	9,6485	10,3616	11,4361
intensity [dB]	neutral	48,9382	68,0121	55,6930	73,3981	77,5032	68,0621	68,0798
	medium	55,2758	69,3792	64,2900	75,3936	78,9833	72,6123	70,6732
	high	63,3062	68,6083	70,3122	76,9817	79,6338	74,9499	74,3547

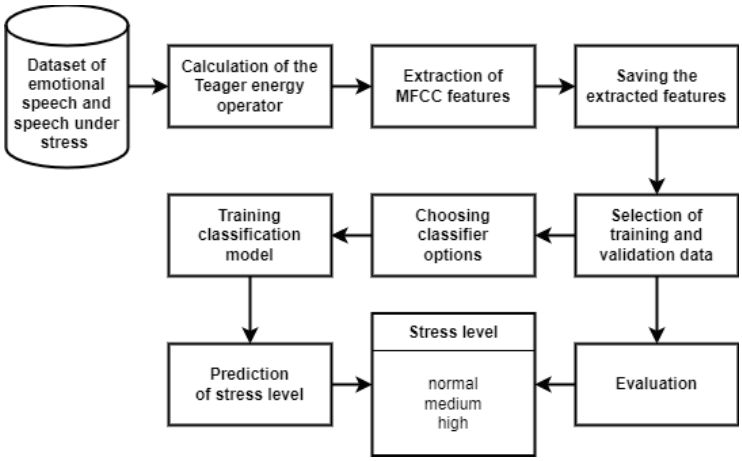

Figure 1. Block scheme of the proposed approach for prediction of level of stress from speech signals

One drawback of conventional decision trees is that produce high-variance estimators. In the case of a small number of training data, observations can dramatically alter the prediction performance of a learned tree.

Bagging (also bootstrap aggregation) is the ensemble learning method that is commonly used to reduce variance within a noisy dataset. Ensemble learning is based on the idea of the 'wisdom of crowds', which suggests that the decision-making of a larger group of is typically better than that of an individual. In bagging, a random sample of data in a training set is selected with replacement, meaning that the individual data points can be chosen more than once. After several data samples are generated, these individual models are then trained independently, and depending on the type of task. Hence each individual tree has high variance, but low bias. Then, averaging the trees reduces the variance, yielding better model performance [20][21].

A. Experimental setup

To implement the proposed approach for classification of speech into individual stress levels (normal, medium, high), we used the MATLAB Classification Learner App[1].

[1] https://www.mathworks.com/help/stats/train-classification-models-in-classification-learner-app.html

When designing our own solution for detecting the presence of stress in speech and classifying the stress level, we were inspired by the research presented in [22].

The whole process of training and classification is depicted in Figure 1. In the first step, a specific dataset of emotional stress speech was loaded. For each speech recording in the dataset, the Teager energy operator (TEO) was first calculated and then the MFCCs were extracted. TEO was proposed for modeling nonlinear speech signals and highlights the frequency and instantaneous changes of the signal amplitude which is very sensitive to subtle changes. MFCCs are a compact representation of the spectrum of a speech signal which describes the overall shape of the spectral envelope. In addition, we also extracted other voice parameters such as Peak2Peak, MaxLocs, MaxWidths, MeanPitch, and so on, to achieve better accuracy of the classifiers. In the next step, we chose the amount of data that was used in the training and validation step. In our case, we used an 80/20 ratio of randomly selected data for training and validation. Subsequently, a suitable classifier was selected, and its parameters were set. As we mentioned earlier, we chose classifiers based on decision trees and ensemble bagged trees. Based on this, a classification model was trained, which was later used for the purpose of classifying speech into three stress levels. Finally, the classification model was evaluated on the validation data.

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TABLE IV.
THE TOTAL NUMBER OF SPEECH UTTERANCES IN THE DATASETS

Number of speech utterances	CRISIS	SUSAS	StressDat	Merged dataset
normal stress	2750	6757	4734	19079
medium stress	2752	3684	3722	11101
high stress	153	3967	3680	8427

TABLE V.
EXPERIMENTAL RESULTS

Accuracy [%]	CRISIS	SUSAS	StressDat	Merged dataset
Decision Tree				
MFCC	71.95	58.21	54.24	56.79
TEO+MFCC	71.62	59.23	52.64	57.40
Bagged Trees				
MFCC	72.24	62.03	**58.30**	**61.83**
TEO+MFCC	**72.59**	**63.48**	54.57	60.23

B. *Results*

We used accuracy to evaluate the classification models. The models were trained separately on the CRISIS, SUSAS, and StressDat datasets, as well as on the merged dataset, which additionally contained emotional speech data from the RAVDESS and SUS datasets that were classified into three stress levels (normal, medium, high).

As can be seen in Table IV, the speech utterances are unevenly distributed among the stress levels. For example, utterances with high stress have the lowest amounts. We selected 8,000 random utterances for each stress level from the merged dataset and trained the classifier on this data to ensure that the classifier does not overtrain.

The results of the prediction and classification of stress levels are shown in Table V. For comparison, we also performed the results when TEO was extracted along with MFCC and when only MFCC was extracted without TEO. The results of the accuracy of the models are somewhat debatable, the influence of the use of TEO is negligible, which may be caused by several factors that affect its calculation and are mainly related to the signal energy. The TEO parameter is mainly used to suppress the noise components and highlight the energy components in the signal. Its use loses its significance in high-quality speech recorded in studio conditions.

We achieved the best overall results in stress prediction and classification using bagged trees. In the case of the CRISIS dataset, we obtained 72.59% classification accuracy, in the case of the SUSAS dataset 63.48%, with the StressDat dataset it was only 58.30%, and on the stress-balanced merged dataset, the accuracy was 61.83%, which are results comparable to other already published research studies, namely [12][15]. Overall, we achieved an average classification accuracy of 64%. Better results could be achieved using classification models based on convolutional neural networks [13].

It should be noted that the classification learner tool only offered us an evaluation using accuracy, more detailed results would be achieved, if we evaluated precision, recall, and F1 score.

VI. CONCLUSION

The main findings of this research can be summarized as follows:

- there exists a large number of emotional speech datasets that can be used to extend the existing collection of datasets containing speech under stress by appropriately mapping emotions into the space of stress levels and thus create large emotional-stress speech datasets;
- a further contribution would be a better understanding of emotional stress by its mapping in the 2D arousal/valence space;
- there are only a few datasets of speech under actual stress, however, public social networks are full of various posts containing recorded stressful situations, these could be used to create such annotated datasets;
- further increase in the dataset size could be achieved by the technique of data augmentation, for example by using generative adversarial networks (GAN);
- fundamental frequency F0 and voice intensity are good predictors of stress levels, on the other hand, using parameters such as jitter, shimmer, or HNR, we cannot clearly determine the presence of stress in a speech;
- the use of the Teager energy operator is relevant only in the case of noisy speech recordings, to reduce the noise and highlight the energy of the signal;
- the use of convolutional neural networks in the process of stress level classification can significantly improve the accuracy of the resulting system [13].

From a commercial use point-of-view, these findings can greatly aid in the research and development of systems to automatically monitor symptoms of acute stress, depression, or mental and emotional states from the voice characteristics. Then, such a system would be used as an auxiliary diagnostic tool or a tool for determining a user's mental or health condition.

Our further research will focus on the extraction and application of speech features other than MFCCs, such as bark frequency cepstral coefficients (BFCCs), gammatone frequency cepstral coefficients (GFCCs), linear frequency cepstral coefficients (LFCCs), and others [23], or their combination; on vectorization and quantization of speech parameters; on the use of convolutional neural networks (CNNs) with dense/fully connected layers in the process of training a classification model; and implementing of end-to-end systems for detecting the presence of stress in human speech.

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REFERENCES

[1] F. Dellaert, T. Polzin, and A. Waibel, "Recognition emotion in speech", in *Proc. of ICSLP*, Philadelphia, PA, USA, 1996, pp. 1970-1973.

[2] J.H.L. Hansen and S.E. Bou-Ghazale, "Getting started with SUSAS: A speech under simulated and actual stress database," in *Proc. of EUROSPEECH*, Rhodes, Greece, 1997.

[3] J. Vamsinath, B. Varshini, T. Sandeep, V. Meghana, and B. Latha, "A survey on stress detection through speech analysis using machine learning," *International Journal of Scientific Research in Science and Technology*, 9(4), 2022, pp. 326-333.

[4] M. Wöllmer, F. Eyben, S. Reiter, B. Schuller, C. Cox, E. Douglas-Cowie, and R. Cowie, "Abandoning emotion classes - Towards continuous emotion recognition with modeling of long-range dependencies," in *Proc. of INTERSPEECH*, Brisbane, Australia, 2008, pp. 597-600.

[5] A. Stuhlsatz, Ch. Meyer, F. Eyben, T. Zielke, G. Meier, B. Schuller, "Deep neural networks for acoustic emotion recognition: Raising the benchmarks," in *Proc. of ICASSP*, Prague, Czech Republic, 2011, pp. 5688-5691.

[6] Q. Mao, M. Dong, Z. Huang, and Y. Zhan, "Learning salient features for speech emotion recognition using convolutional neural networks," *IEEE Transactions on Multimedia*, 16(8), 2014.

[7] G. Trigeorgis, F. Ringeval, R. Brueckner, E. Marchi, M.A. Nicolaou, B, Schuller, and S. Zafeiriou, "Adieu features? End-to-end speech emotion recognition using a deep convolutional recurrent network," in *Proc. of ICASSP*, Shanghai, China, 2016.

[8] B.B. Rad and M. Moradhaseli, "Speech emotion recognition methods: A literature review," *AIP Conference Proceedings*, 1891(020105), 2017.

[9] S. Zafeiriou, D. Kollias, M.A. Nicolaou, A. Papaioannou, G. Zhao, I. Kotsia, "Aff-Wild: Valence and arousal 'in-the-wild' challenge," in *Proc. of the 30th IEEE Conference on Computer Vision and Pattern Recognition*, Honolulu, Hawaii, USA, 2016.

[10] G. Demenko and M. Jastrzebska, "Analysis of natural speech under stress," *Acta Physica Polonica Series A*, 121(1-A), 2012, pp. A92-A95.

[11] M. van Puyvelde, X. Neyt, F. McGlone, and N. Pattyn, "Voice stress analysis: A new framework for voice and effort in human performance," *Frontiers in Psychology*, 9, 2018, Article 1994.

[12] H. Han, K. Byun, and H.-G. Kang, "A deep learning-based stress detection algorithm with speech signal," in *Proc. of AVSU: Workshop on Audio-Visual Scene Understanding for Immersive Multimedia*, Seoul, Republic of Korea, 2018, pp. 11-15.

[13] S. Vaikole, S. Mulajkar, A. More, P. Jayaswal, and S. Dhas, "Stress detection through speech analysis using machine learning," *International Journal of Creative Research Thoughts*, 8(5), 2020.

[14] S.R. Livingstone and F.A. Russo, "The Ryerson audio-visual database of emotional speech and song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in North American English," *PLoS ONE*, 13(5), 2018, e0196391.

[15] M.S.H. Hilmy, A.L. Asnawi, A.Z. Jusoh, K. Abdullah, S.N. Ibrahim, H.A.M. Ramli, and N.F.M. Azmin, "Stress classification based on speech analysis of MFCC feature via machine learning," in *Proc. of the 8th International Conference on Computer and Communication Engineering* (ICCCE), Kuala Lumpur, Malaysia, 2021, pp. 339-343.

[16] R. Sabo, J. Rajčáni, and M. Ritomský, "Designing database of speech under stress using a simulation in virtual reality," in *Proc. of DISA 2018 – IEEE World Symposium on Digital Intelligence for Systems and Machines*, Košice, Slovakia, 2018, pp. 321-326.

[17] L. Macková, A. Čižmár, and J. Juhár, "A study of acoustic features for emotional speaker recognition in i-vector representation," *Acta Electrotechnica et Informatica*, 15(2), 2015, pp. 15-20.

[18] M. Rusko, S. Darjaa, M. Trnka, R. Sabo, and M. Ritomský, "Expressive speech synthesis for critical situations," *Computing and Informatics*, 33(6), 2014, pp. 1312-1332.

[19] R. Sabo, Š. Beňuš, M. Trnka, M. Ritomský, M. Rusko, M. Schaper, and J. Szabo, "StressDat – Database of speech under stress in Slovak," *Journal of Linguistics*, 72(2), 2021, pp. 579-589.

[20] D. Opitz and R. Maclin, "Popular ensemble methods: An empirical study," *Journal of Artificial Intelligence Research*, 11(1999), 1999, pp. 169-198.

[21] Breiman, L., "Random forests," *Machine Learning*, 45, 2001, 33 pages.

[22] S.R. Bandela and T.K. Kumar, "Stressed speech emotion recognition using feature fusion of Teager energy operator and MFCC," in *Proc. of the 8th International Conference on Computing, Communication and Networking Technologies* (ICCCNT), Delhi, India, 2017, pp 1-5.

[23] V.V. Yerigeri and L.K. Ragha, "Speech stress recognition using semi-eager learning," *Cognitive Systems Research*, 65, 2021, pp. 79-97.

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